

Machine Learning–based Late Delivery Risk Prediction in Global Supply Chain Operations

A Research Project Report

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Palak Bhavsar

ABSTRACT

The rapid growth of e-commerce and global supply chain operations has significantly increased the importance of efficient logistics and timely deliveries. Delivery delays can negatively affect customer satisfaction, increase operational costs, and impact the overall performance of logistics organizations. Predicting potential delivery delays in advance enables companies to take proactive measures, optimize resource allocation, and improve operational efficiency. Machine learning provides an effective approach for identifying patterns in historical logistics data and forecasting delivery risks with high accuracy.

This research presents a machine learning-based solution for predicting late delivery risk using the APL Logistics dataset. The dataset contains various attributes related to customer information, order details, shipping methods, product categories, sales values, and geographical locations. The raw dataset was preprocessed through data cleaning, handling of missing values, removal of duplicate records, feature engineering, label encoding of categorical variables, and feature scaling to improve model performance and data quality.

The processed dataset was used to train an XGBoost Classifier, selected for its ability to handle structured data efficiently and deliver high predictive performance. The model was developed to classify whether an order is likely to be delivered late based on historical shipping patterns and order characteristics. Model performance was evaluated using standard classification metrics and feature importance analysis to understand the influence of different variables on prediction outcomes.

To make the model practical for business users, an interactive dashboard was developed using Streamlit. The dashboard provides multiple modules, including an Executive Dashboard, Risk Analytics, Regional Analytics, a Prediction Center for real-time risk prediction, and a Model Performance page. These modules allow users to explore logistics data, monitor delivery trends, visualize operational insights, and generate predictions through a user-friendly interface.

The proposed system demonstrates how machine learning can support logistics decision-making by identifying shipments that are at risk of delayed delivery before they occur. Such predictive capabilities can help organizations improve supply chain planning, reduce delivery delays, optimize shipping strategies, and enhance customer satisfaction. The project highlights the practical application of data science and machine learning in solving real-world logistics challenges through predictive analytics and interactive visualization.

Keywords: Machine Learning, XGBoost, Logistics, Late Delivery Prediction, Predictive Analytics, Supply Chain Management, Streamlit, Data Visualization, Classification.

CHAPTER 1: INTRODUCTION

1.1 Background

The logistics and supply chain industry has become one of the most important sectors supporting global trade and e-commerce. With the rapid growth of online shopping and international business operations, organizations are expected to deliver products quickly, efficiently, and accurately. Customers now consider timely delivery as one of the primary indicators of service quality, making delivery performance a critical factor for business success. However, various operational challenges such as transportation delays, inefficient route planning, weather conditions, inventory issues, and increasing order volumes often result in late deliveries.

Late deliveries negatively affect both customers and organizations. Customers may lose confidence in a company's services, leading to reduced customer satisfaction and loyalty. For businesses, delayed shipments increase operational costs, reduce supply chain efficiency, and negatively impact overall profitability. Therefore, organizations are increasingly adopting data-driven approaches to predict and prevent delivery delays before they occur.

Machine Learning has emerged as an effective technology for analyzing large volumes of logistics data and identifying hidden patterns that are difficult to detect using traditional statistical methods. By learning from historical delivery records, machine learning algorithms can predict the probability of delayed deliveries, allowing logistics managers to take preventive actions such as modifying shipping routes, reallocating resources, or selecting alternative transportation methods.

The APL Logistics dataset contains valuable information related to customer demographics, product details, shipping methods, sales information, geographical regions, and order characteristics. These attributes provide sufficient information to build predictive models capable of estimating the likelihood of delivery delays.

In this research, an XGBoost Classifier is used to develop a predictive model for late delivery risk. XGBoost is a powerful ensemble learning algorithm that provides excellent performance on structured datasets, handles missing values efficiently, and offers feature importance analysis. To improve the usability of the prediction system, an interactive dashboard is developed using Streamlit, enabling users to visualize logistics data, explore operational insights, and generate real-time delivery risk predictions.

This research demonstrates how machine learning and data visualization can be integrated to support better decision-making in logistics operations, ultimately helping organizations improve delivery performance and customer satisfaction.

1.2 Problem Statement

Timely delivery is one of the most significant challenges in modern logistics operations. Despite advancements in transportation technologies and supply chain management, many

organizations continue to experience delivery delays due to multiple operational and environmental factors. Traditional monitoring systems mainly identify delays after they occur, providing limited opportunities for preventive action.

The absence of an intelligent prediction system prevents logistics managers from identifying high-risk shipments in advance. As a result, organizations face increased operational costs, inefficient resource allocation, delayed customer deliveries, and reduced customer satisfaction.

The primary problem addressed in this research is the development of a machine learning-based system capable of predicting late delivery risk before shipment completion. Such a system can assist logistics companies in making proactive decisions that minimize delays and improve overall operational efficiency.

1.3 Objectives of the Study

The primary objective of this research is to develop a machine learning-based predictive system capable of identifying shipments that are likely to experience delivery delays.

The specific objectives include:

- To analyze the APL Logistics dataset and understand the factors affecting delivery performance.
- To perform data preprocessing, cleaning, and feature engineering to improve data quality.
- To develop a predictive model using the XGBoost Classifier.
- To evaluate the performance of the developed machine learning model using classification metrics.
- To identify the most influential factors contributing to late deliveries through feature importance analysis.
- To design and develop an interactive Streamlit dashboard for visualization and prediction.
- To provide business insights that support operational decision-making and delivery risk management.

1.4 Scope of the Study

The scope of this research is limited to predicting late delivery risk using historical logistics data available in the APL Logistics dataset. The study focuses on supervised machine learning techniques for binary classification, where each shipment is classified as either likely to experience a delayed delivery or not.

The research includes data preprocessing, feature engineering, machine learning model development, performance evaluation, and dashboard implementation. The developed system is intended as a decision-support tool for logistics professionals and demonstrates how predictive analytics can improve operational planning.

This study does not include real-time GPS tracking, route optimization algorithms, live weather integration, or dynamic transportation scheduling. The prediction model relies solely on historical data available within the dataset.

1.5 Research Methodology

The research follows a structured methodology consisting of several sequential phases.

Initially, the APL Logistics dataset is collected and examined to understand its structure and quality. Data preprocessing techniques such as handling missing values, removing duplicate records, feature engineering, label encoding, and feature scaling are applied to prepare the dataset for machine learning.

The processed data is divided into training and testing sets. The XGBoost Classifier is then trained using the prepared dataset to predict late delivery risk. Model performance is evaluated using standard classification metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis.

Finally, an interactive dashboard is developed using Streamlit to present business insights, visualize logistics data, and allow users to generate delivery risk predictions through a simple interface.

1.6 Expected Outcomes

The expected outcome of this research is the development of an intelligent prediction system capable of accurately identifying shipments that are at risk of delayed delivery.

The developed model is expected to:

- Improve the prediction of late delivery risk.
- Support logistics managers in proactive decision-making.
- Enhance supply chain visibility through interactive visualizations.
- Reduce operational risks associated with delayed shipments.
- Demonstrate the practical application of machine learning in logistics and supply chain management.

Furthermore, the Streamlit dashboard is expected to provide a user-friendly interface that enables non-technical users to interact with the predictive model and gain meaningful business insights from logistics data.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The rapid advancement of digital technologies has transformed the logistics and supply chain industry. Organizations generate large volumes of operational data from customer transactions, transportation systems, inventory management, and order processing. This data provides valuable insights that can be utilized through data analytics and machine learning to improve business performance. Among the many challenges faced by logistics companies, predicting delivery delays remains one of the most important because delayed shipments directly affect customer satisfaction, operational efficiency, and overall profitability.

Traditional logistics systems primarily depend on historical reports and manual monitoring to identify delayed shipments. While these approaches help organizations understand past performance, they often fail to predict future delivery risks in advance. Recent research has therefore focused on applying machine learning techniques that can learn complex patterns from historical logistics data and provide early predictions of delivery delays.

This chapter reviews previous studies related to logistics analytics, delivery delay prediction, machine learning classification techniques, and interactive business intelligence dashboards. It also identifies the research gap that motivated the development of the proposed system.

2.2 Machine Learning in Logistics and Supply Chain Management

Machine learning has become one of the most widely adopted technologies in modern supply chain management. It enables organizations to analyze large datasets, identify hidden patterns, and make intelligent predictions that improve operational efficiency. Logistics companies use machine learning for demand forecasting, route optimization, warehouse automation, inventory management, fraud detection, and delivery risk prediction.

Predictive models analyze historical shipment records to estimate whether future deliveries are likely to experience delays. Factors such as shipping mode, order quantity, product category, geographical location, customer segment, seasonal demand, and transportation routes significantly influence delivery performance.

The increasing availability of structured logistics data has encouraged researchers to apply supervised learning algorithms that classify deliveries into on-time and delayed categories. These predictive systems enable organizations to take preventive actions before delivery failures occur, reducing operational costs and improving customer satisfaction.

2.3 Previous Studies on Delivery Delay Prediction

Several researchers have explored the application of predictive analytics in logistics. Early studies mainly relied on statistical methods such as linear regression and probability-based

forecasting models. Although these techniques provided basic insights, they struggled to capture complex relationships among multiple variables.

Recent research has shifted towards machine learning algorithms because of their superior predictive capabilities. Decision Trees, Random Forest, Support Vector Machines (SVM), Artificial Neural Networks, Gradient Boosting, and XGBoost have been extensively used to classify delivery risks. These models are capable of handling nonlinear relationships, high-dimensional datasets, and complex interactions among variables.

Studies have reported that ensemble learning algorithms generally outperform individual classifiers because they combine multiple decision trees to improve prediction accuracy while reducing overfitting. Among these methods, XGBoost has consistently demonstrated strong performance due to its efficiency, scalability, and ability to manage missing values effectively.

Researchers have also emphasized the importance of feature engineering in improving model performance. Creating meaningful variables from existing data often enhances prediction accuracy and provides better business insights.

2.4 XGBoost for Predictive Analytics

Extreme Gradient Boosting (XGBoost) is one of the most successful machine learning algorithms used for structured datasets. It belongs to the family of gradient boosting algorithms and builds multiple decision trees sequentially, where each tree attempts to correct the errors of previous trees.

Compared to traditional boosting methods, XGBoost offers several advantages:

- High prediction accuracy
- Fast training speed
- Efficient handling of missing values
- Regularization techniques that reduce overfitting
- Parallel processing capabilities
- Feature importance analysis
- Scalability for large datasets

Due to these advantages, XGBoost has become a preferred choice for predictive analytics in finance, healthcare, manufacturing, retail, and logistics. In this research, XGBoost was selected because it provides reliable performance for binary classification problems involving structured logistics data.

2.5 Data Visualization and Business Intelligence

Predictive models are most valuable when their results can be easily interpreted by business users. Interactive dashboards have therefore become an essential component of modern data analytics systems. Business Intelligence (BI) tools enable managers to visualize operational performance through graphs, charts, and real-time analytics.

Streamlit has emerged as a popular Python framework for developing machine learning dashboards because of its simplicity and ability to integrate predictive models with interactive visualizations. Unlike traditional reporting systems, Streamlit applications allow users to interact with datasets, generate predictions, and explore insights without requiring programming knowledge.

In logistics management, interactive dashboards assist decision-makers by presenting shipment statistics, regional performance, sales trends, customer analysis, and delivery risk predictions in an intuitive format.

2.6 Research Gap

Although previous studies have demonstrated the effectiveness of machine learning in logistics, several limitations remain.

Many existing studies primarily focus on developing predictive models without providing an integrated visualization platform for end users. Others evaluate multiple algorithms but do not include feature engineering techniques that improve business interpretation. Some systems rely on static reports, limiting real-time interaction and decision support.

Furthermore, relatively few studies combine advanced machine learning algorithms such as XGBoost with an interactive dashboard that allows users to visualize logistics performance and generate real-time predictions from the same application.

The present research addresses these limitations by integrating data preprocessing, feature engineering, machine learning, model evaluation, and interactive dashboard development into a single predictive analytics system. This approach provides both accurate delivery risk prediction and a practical decision-support interface for logistics management.

2.7 Summary

The literature review demonstrates that machine learning has significantly improved predictive analytics in logistics and supply chain management. Among various algorithms, XGBoost offers high prediction accuracy and computational efficiency for structured logistics datasets. Previous research also highlights the growing importance of interactive dashboards for translating predictive insights into practical business decisions.

Building upon these findings, the present research develops an integrated system that combines data preprocessing, feature engineering, machine learning, and Streamlit-based visualization to predict late delivery risk effectively. The proposed solution contributes to improving operational planning, reducing delivery delays, and enhancing customer satisfaction through data-driven decision-making.

CHAPTER 3: DATASET DESCRIPTION AND EXPLORATORY DATA ANALYSIS

3.1 Introduction

The success of any machine learning model largely depends on the quality and characteristics of the dataset used for training and evaluation. In predictive analytics, understanding the structure, attributes, and relationships within the data is essential for developing accurate and reliable models. This chapter presents a detailed description of the dataset used in this research and discusses the exploratory data analysis (EDA) performed before data preprocessing and model development.

The dataset provides comprehensive information related to logistics operations, customer details, order transactions, shipping methods, product information, and delivery performance. Through exploratory analysis, valuable insights were obtained regarding the distribution of variables, potential data quality issues, and factors influencing late deliveries.

df.head()

	Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Category Id	Category Name	Customer City	Customer Country	Customer Fname	Customer Id	Customer Lname	Customer Segmen
0	DEBIT	6	4	159.69	472.45	Late delivery	1	9	Cardio Equipment	Brownsville	EE. UU.	Richard	1	Hernandez	Consum
1	DEBIT	4	4	48.71	167.96	Shipping on time	0	29	Shop By Sport	Littleton	EE. UU.	Mary	2	Barrett	Consum
2	DEBIT	4	4	87.36	181.99	Shipping on time	0	48	Water Sports	Littleton	EE. UU.	Mary	2	Barrett	Consum
3	DEBIT	6	4	-41.89	175.99	Late delivery	1	48	Water Sports	Littleton	EE. UU.	Mary	2	Barrett	Consum
4	DEBIT	6	4	10.00	40.00	Late delivery	1	24	Women's Apparel	Littleton	EE. UU.	Mary	2	Barrett	Consum

3.2 Dataset Overview

The research utilizes the **APL Logistics dataset**, which contains historical shipment and order information collected from logistics operations. The dataset consists of transactional records representing customer orders along with information about products, shipping details, geographical locations, sales, and delivery status.

Each record corresponds to an individual customer order and contains both numerical and categorical attributes that describe different aspects of the shipment process. The dataset is suitable for supervised machine learning because it includes a target variable indicating whether an order experienced a delayed delivery.

The dataset was imported into Python using the Pandas library and stored as a DataFrame for further analysis and preprocessing.

```
[9]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180519 entries, 0 to 180518
Data columns (total 40 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   Type                                   180519 non-null  object
 1   Days for shipping (real)               180519 non-null  int64
 2   Days for shipment (scheduled)          180519 non-null  int64
 3   Benefit per order                     180519 non-null  float64
 4   Sales per customer                    180519 non-null  float64
 5   Delivery Status                       180519 non-null  object
 6   Late_delivery_risk                    180519 non-null  int64
 7   Category Id                           180519 non-null  int64
 8   Category Name                         180519 non-null  object
 9   Customer City                         180519 non-null  object
10  Customer Country                      180519 non-null  object
11  Customer Fname                        180519 non-null  object
12  Customer Id                           180519 non-null  int64
13  Customer Lname                        180511 non-null  object
14  Customer Segment                      180519 non-null  object
15  Customer State                        180519 non-null  object
16  Customer Street                       180519 non-null  object
17  Customer Zipcode                      180516 non-null  float64
18  Department Id                         180519 non-null  int64
19  Department Name                       180519 non-null  object
20  Latitude                              180519 non-null  float64
21  Longitude                             180519 non-null  float64
22  Market                                180519 non-null  object
23  Order City                            180519 non-null  object
24  Order Country                         180519 non-null  object
25  Order Customer Id                    180519 non-null  int64
26  Order Item Discount                  180519 non-null  float64
27  Order Item Discount Rate             180519 non-null  float64
28  Order Item Product Price             180519 non-null  float64
29  Order Item Profit Ratio              180519 non-null  float64
30  Order Item Quantity                  180519 non-null  int64
31  Sales                                180519 non-null  float64
32  Order Item Total                     180519 non-null  float64
33  Order Profit Per Order               180519 non-null  float64
34  Order Region                         180519 non-null  object
35  Order State                          180519 non-null  object
36  Order Status                         180519 non-null  object
37  Product Name                         180519 non-null  object
38  Product Price                        180519 non-null  float64
39  Shipping Mode                        180519 non-null  object
dtypes: float64(13), int64(8), object(19)
memory usage: 55.1+ MB
```

3.3 Dataset Characteristics

The dataset contains information from multiple operational areas within the logistics process.

These include:

- Customer Information
- Product Information
- Order Details
- Shipping Information
- Sales Information
- Geographic Information

- Delivery Performance

The target variable used for classification is **Late_delivery_risk**, which indicates whether an order is likely to be delivered late.

The dataset includes both **numerical** and **categorical** features, making it suitable for classification algorithms after appropriate preprocessing techniques such as label encoding and feature scaling.

```
df.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Days for shipping (real)	180519.0	3.497654	1.623722	0.000000	2.000000	3.000000	5.000000	6.000000
Days for shipment (scheduled)	180519.0	2.931847	1.374449	0.000000	2.000000	4.000000	4.000000	4.000000
Benefit per order	180519.0	21.974989	104.433526	-4274.980000	7.000000	31.520000	64.800000	911.800000
Sales per customer	180519.0	183.107607	120.043668	7.490000	104.380000	163.990000	247.400000	1939.990000
Late_delivery_risk	180519.0	0.548291	0.497664	0.000000	0.000000	1.000000	1.000000	1.000000
Category Id	180519.0	31.851451	15.640064	2.000000	18.000000	29.000000	45.000000	76.000000
Customer Id	180519.0	6691.379495	4162.918106	1.000000	3258.500000	6457.000000	9779.000000	20757.000000
Customer Zipcode	180516.0	35921.126914	37542.461122	603.000000	725.000000	19380.000000	78207.000000	99205.000000
Department Id	180519.0	5.443460	1.629246	2.000000	4.000000	5.000000	7.000000	12.000000
Latitude	180519.0	29.719955	9.813646	-33.937553	18.265432	33.144863	39.279617	48.781933
Longitude	180519.0	-84.915675	21.433241	-158.025986	-98.446312	-76.847908	-66.370583	115.263077
Order Customer Id	180519.0	6691.379495	4162.918106	1.000000	3258.500000	6457.000000	9779.000000	20757.000000
Order Item Discount	180519.0	20.664741	21.800901	0.000000	5.400000	14.000000	29.990000	500.000000
Order Item Discount Rate	180519.0	0.101668	0.070415	0.000000	0.040000	0.100000	0.160000	0.250000
Order Item Product Price	180519.0	141.232547	139.732489	9.990000	50.000000	59.990000	199.990000	1999.990000
Order Item Profit Ratio	180519.0	0.120647	0.466796	-2.750000	0.080000	0.270000	0.360000	0.500000
Order Item Quantity	180519.0	2.127638	1.453451	1.000000	1.000000	1.000000	3.000000	5.000000
Sales	180519.0	203.772092	132.273075	9.990000	119.980000	199.920000	299.950000	1999.990000
Order Item Total	180519.0	183.107607	120.043668	7.490000	104.380000	163.990000	247.400000	1939.990000
Order Profit Per Order	180519.0	21.974989	104.433526	-4274.980000	7.000000	31.520000	64.800000	911.800000
Product Price	180519.0	141.232547	139.732489	9.990000	50.000000	59.990000	199.990000	1999.990000

3.4 Feature Categories

To simplify analysis, the dataset features can be grouped into several logical categories.

Customer Features

These variables describe customer-related information.

Examples include:

- Customer Segment
- Customer City
- Customer State
- Customer Country
- Customer Zipcode

These features help identify customer demographics and regional purchasing behavior.

Product Features

These variables provide information regarding purchased products.

Examples include:

- Product Price
- Category Name
- Category ID
- Department Name
- Department ID

These variables are useful for understanding how different product categories influence delivery performance.

Order Features

Order-related variables describe transaction characteristics.

Examples include:

- Sales
- Sales per Customer
- Order Item Quantity
- Order Item Discount
- Order Item Total
- Benefit per Order
- Order Profit Per Order

These features capture financial and operational aspects of each customer order.

Shipping Features

Shipping variables represent transportation-related information.

Examples include:

- Shipping Mode

- Scheduled Shipping Days
- Shipping Pressure Index
- Shipping Efficiency
- Shipping Mode Risk

These variables directly influence delivery timelines and play a significant role in predicting delayed deliveries.

Geographic Features

Geographical information includes:

- Order Country
- Order Region
- Order State
- Order City
- Market

These variables help analyze delivery performance across different locations and markets.



3.5 Target Variable

The dependent variable used in this research is:

Late_delivery_risk

This variable represents whether an order is expected to experience delayed delivery.

The classification labels are:

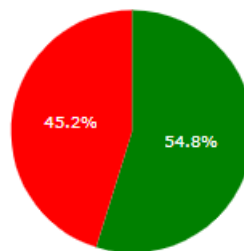
Value Meaning

0 No Late Delivery

1 Late Delivery

The objective of the machine learning model is to predict this binary outcome using the available order, shipping, customer, and product information.

Late Delivery Risk Distribution



3.6 Exploratory Data Analysis (EDA)

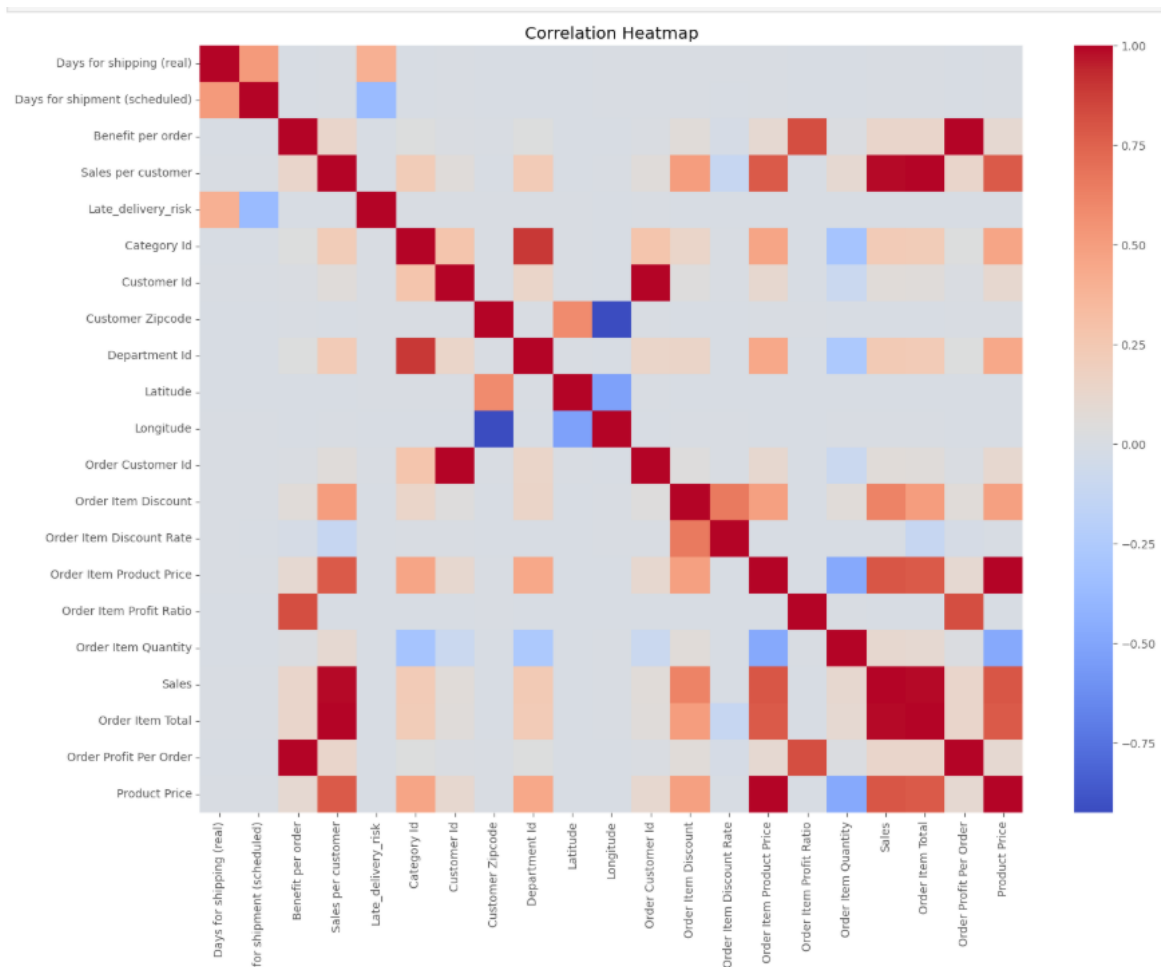
Exploratory Data Analysis (EDA) was performed to understand the dataset before applying machine learning algorithms. The analysis focused on identifying missing values, duplicate records, variable distributions, and relationships among different features.

The following activities were carried out during EDA:

- Inspection of dataset dimensions
- Data type analysis
- Summary statistics
- Missing value identification
- Duplicate record detection

- Distribution analysis of numerical variables
- Frequency analysis of categorical variables
- Correlation analysis
- Target variable distribution

These analyses provided valuable insights into the structure and quality of the dataset and guided the preprocessing stage.



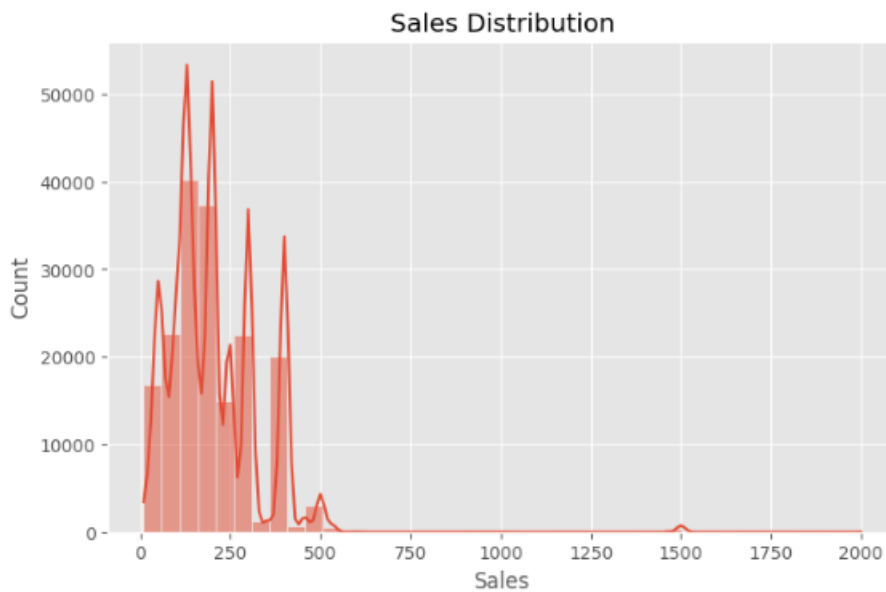
3.7 Key Insights from EDA

The exploratory analysis revealed several important observations:

- The dataset contained both numerical and categorical variables requiring different preprocessing techniques.
- Certain customer and product attributes showed significant variation across different geographical regions.
- Shipping-related variables exhibited strong relationships with delivery outcomes.

- Sales values and order quantities displayed a wide range of distributions, indicating the presence of both small and large transactions.
- Feature engineering helped derive additional business-oriented variables that improved the predictive capability of the machine learning model.
- The target variable demonstrated sufficient class representation for developing a binary classification model.

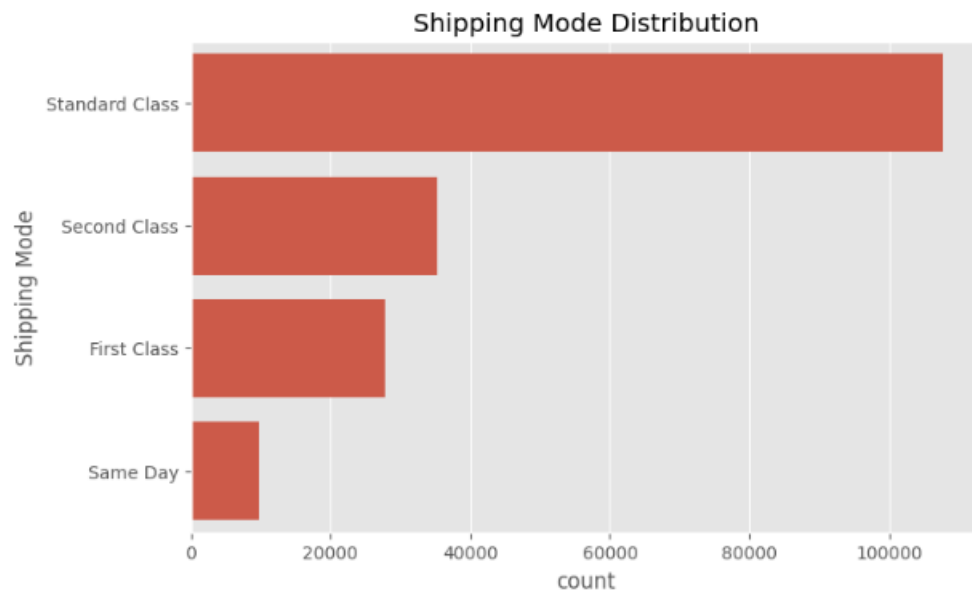
These findings supported the selection of relevant features and informed the subsequent preprocessing steps.



3.8 Importance of Data Exploration

Exploratory Data Analysis is an essential step in machine learning because it improves understanding of the dataset before model development. Proper exploration helps identify inconsistencies, detect outliers, understand feature relationships, and reduce the risk of developing biased or inaccurate models.

In this research, EDA served as the foundation for feature engineering, encoding categorical variables, scaling numerical features, and selecting the most relevant variables for training the XGBoost classifier.



3.9 Chapter Summary

This chapter introduced the APL Logistics dataset and explained its structure, feature categories, and target variable. Exploratory Data Analysis provided important insights into customer behavior, shipping characteristics, sales patterns, and delivery performance. The knowledge gained from EDA guided the preprocessing process and contributed to the development of an effective machine learning model for predicting late delivery risk.

CHAPTER 4: DATA PREPROCESSING AND FEATURE ENGINEERING

4.1 Introduction

Data preprocessing is a fundamental step in the machine learning pipeline, as the quality of the input data directly influences the performance of predictive models. Real-world datasets often contain inconsistencies such as missing values, duplicate records, irrelevant attributes, and categorical variables that cannot be processed directly by machine learning algorithms. Therefore, preprocessing is essential to improve data quality and ensure reliable model training.

In this research, the APL Logistics dataset underwent several preprocessing steps, including data inspection, handling missing values, removing duplicate records, eliminating data leakage, feature engineering, categorical data encoding, feature scaling, and train-test splitting. These operations ensured that the dataset was clean, consistent, and suitable for building an accurate late delivery prediction model.

4.2 Data Inspection

The preprocessing process began with a detailed inspection of the dataset to understand its structure and identify potential quality issues. The dataset was imported into Python using the Pandas library and stored as a DataFrame.

The following operations were performed:

- Displaying the first few records of the dataset
- Examining column names and data types
- Identifying numerical and categorical variables
- Computing summary statistics
- Determining the number of records and features

This initial inspection provided a clear understanding of the dataset and guided the subsequent preprocessing steps.

4.3 Handling Missing Values

Missing values are common in logistics datasets due to incomplete records or errors during data collection. Such missing information can negatively affect machine learning algorithms by introducing bias and reducing prediction accuracy.

The dataset was examined for missing values using Pandas functions. Variables with missing entries were carefully analyzed before appropriate handling techniques were applied. Missing values were either removed or treated according to the characteristics of the respective feature.

The preprocessing process ensured that the final dataset used for model training contained complete and consistent information.

4.4 Duplicate Record Removal

Duplicate records increase redundancy and may bias predictive models by giving excessive importance to repeated observations. Therefore, duplicate entries were identified using the `duplicated()` function.

Any duplicate records detected within the dataset were removed before model development. This improved the quality of the dataset and ensured that each observation represented a unique logistics transaction.

4.5 Removal of Data Leakage

Data leakage occurs when variables provide direct information about the target variable, leading to unrealistically high prediction accuracy. Such variables are unavailable before prediction and therefore cannot be used in a practical machine learning system.

To prevent leakage, delivery outcome variables that reveal the actual delivery status were removed before training the model. Examples include:

- Delivery Status
- Days for Shipping (Real)
- Delivery Gap
- Delivery Performance

Removing these variables ensured that the model learned meaningful relationships from available order and shipping information rather than relying on outcome-related attributes.

4.6 Feature Engineering

Feature engineering was performed to create additional variables that better represent business characteristics and improve predictive performance.

Several new features were derived from existing variables, including:

- Shipping Pressure Index
- Discount Percentage
- Order Value Category
- Profit Category

- High Value Customer
- Express Shipping
- Large Order
- Profit Margin
- Sales Category
- Customer Value Score
- Shipping Efficiency
- Order Complexity Score
- Market Risk Score
- Shipping Mode Risk

These engineered variables capture hidden relationships within the logistics data and provide additional information to the machine learning model.

Feature engineering significantly enhanced the predictive capability of the XGBoost classifier by incorporating meaningful business indicators rather than relying solely on raw transactional data.

4.7 Encoding Categorical Variables

Machine learning algorithms require numerical input for model training. However, the logistics dataset contained numerous categorical variables such as shipping mode, customer segment, product category, market, and geographical information.

These categorical attributes were transformed into numerical values using Label Encoding. Each unique category was assigned a corresponding numerical identifier while maintaining consistency across training and prediction phases.

The trained label encoders were saved using the Joblib library and reused during dashboard prediction to ensure consistent encoding of user-selected records.

4.8 Feature Scaling

The dataset contains numerical variables with different units and value ranges. For example, sales values are measured in currency, while discounts are represented as percentages and quantities are measured as integers.

To improve model consistency and maintain standardized feature values, numerical attributes were normalized using the StandardScaler technique.

Feature scaling ensures that variables with larger numerical ranges do not dominate the learning process and contributes to stable model performance.

The fitted scaler object was stored using Joblib and later loaded into the Streamlit dashboard to preprocess new prediction inputs consistently.

4.9 Train-Test Split

To evaluate model performance objectively, the processed dataset was divided into training and testing subsets.

A train-test ratio of 80:20 was adopted, where:

- 80% of the data was used for model training.
- 20% of the data was reserved for model testing.

The split was performed using the `train_test_split()` function from Scikit-learn with a fixed random state to ensure reproducibility of experimental results.

This approach enables the model to learn from historical data while being evaluated on previously unseen observations.

4.10 Preprocessed Dataset

After completing all preprocessing operations, the final dataset consisted of cleaned, encoded, engineered, and scaled features suitable for supervised machine learning.

The resulting dataset exhibited the following characteristics:

- No duplicate records
- Clean feature set
- Encoded categorical variables
- Standardized numerical features
- Additional engineered business features
- Balanced training and testing datasets

This prepared dataset served as the input for training the XGBoost classification model described in the next chapter.

4.11 Chapter Summary

This chapter described the preprocessing pipeline applied to the APL Logistics dataset before machine learning model development. Data inspection, handling of missing values, duplicate

removal, prevention of data leakage, feature engineering, label encoding, feature scaling, and train-test splitting collectively improved the quality and usability of the dataset. The resulting processed dataset formed a strong foundation for developing an accurate and reliable late delivery prediction model.

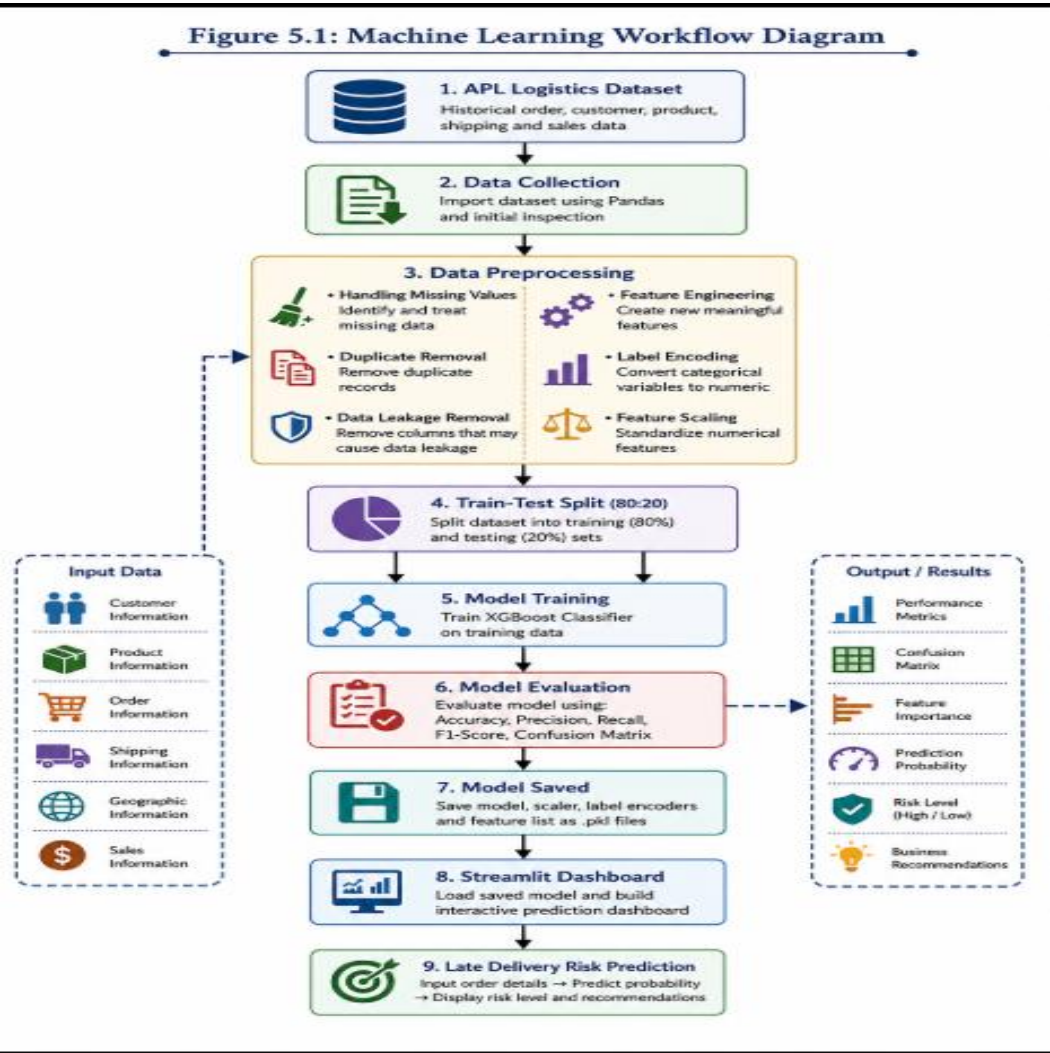
CHAPTER 5: MACHINE LEARNING MODEL DEVELOPMENT

5.1 Introduction

Machine learning plays a significant role in predictive analytics by enabling systems to learn patterns from historical data and make accurate predictions on unseen observations. In logistics and supply chain management, predictive models can help organizations identify potential delivery delays before they occur, allowing timely corrective actions and improved operational planning.

The primary objective of this research is to develop a binary classification model capable of predicting whether an order is at risk of being delivered late. After preprocessing and feature engineering, the prepared dataset was used to train a machine learning model. Among various supervised learning algorithms, the **Extreme Gradient Boosting (XGBoost) Classifier** was selected because of its excellent predictive performance, computational efficiency, and ability to handle structured datasets with both numerical and categorical features.

This chapter discusses the algorithm selection, model architecture, training methodology, evaluation techniques, and performance of the developed prediction model.



5.2 Selection of Machine Learning Algorithm

Selecting an appropriate machine learning algorithm is one of the most important decisions in predictive modeling. Several classification algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Gradient Boosting can be used for binary classification problems. However, each algorithm has different strengths and limitations depending on the characteristics of the dataset.

For this research, the **XGBoost Classifier** was selected because it provides high prediction accuracy while maintaining computational efficiency. XGBoost is an ensemble learning algorithm based on gradient boosting, where multiple decision trees are built sequentially. Each successive tree attempts to reduce the prediction errors made by the previous trees, resulting in a stronger overall model.

The algorithm offers several advantages that make it suitable for logistics prediction tasks:

- High predictive accuracy on structured datasets.
- Efficient handling of numerical and categorical variables after encoding.
- Regularization techniques that reduce overfitting.
- Built-in feature importance analysis.
- Faster training through parallel computation.
- Scalability for large datasets.

These characteristics make XGBoost particularly effective for predicting late delivery risk in logistics operations.

5.3 Model Architecture

The developed prediction model follows a supervised learning approach. The input consists of processed logistics records containing customer information, order characteristics, product details, geographical information, shipping attributes, and engineered business features.

The target variable is **Late_delivery_risk**, which represents whether a shipment is expected to experience delayed delivery.

The model learns relationships between the independent variables and the target variable during the training phase. Once trained, it predicts the probability of late delivery for new shipment records.

The overall machine learning workflow consists of:

1. Data preprocessing
2. Feature engineering

3. Label encoding
4. Feature scaling
5. Train-test split
6. XGBoost model training
7. Model evaluation
8. Prediction generation
9. Dashboard integration

This pipeline ensures consistency between model development and real-time prediction in the Streamlit dashboard.

5.4 Model Training

The prepared dataset was divided into training and testing subsets using an 80:20 ratio. The training dataset was used to learn the underlying relationships between logistics features and delivery outcomes, while the testing dataset was reserved for evaluating model performance on previously unseen data.

The XGBoost Classifier was trained using the processed training dataset. During training, the algorithm constructed multiple decision trees iteratively, with each tree improving upon the errors of the previous one. This boosting mechanism enabled the model to capture complex nonlinear relationships among variables while minimizing prediction errors.

The trained model was saved using the **Joblib** library, allowing it to be reused within the Streamlit dashboard for real-time predictions without requiring retraining.

5.5 Model Evaluation

Evaluating a machine learning model is essential to determine its reliability and predictive capability. Several standard classification metrics were used to assess the performance of the developed XGBoost model.

Accuracy

Accuracy measures the proportion of correctly classified observations among all predictions made by the model.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

A higher accuracy indicates better overall predictive performance.

Precision

Precision measures the proportion of correctly predicted delayed deliveries among all deliveries predicted as delayed.

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision reduces false alarms and improves decision-making.

Recall

Recall measures the model's ability to correctly identify actual delayed deliveries.

$$\text{Recall} = \frac{TP}{TP + FN}$$

A high recall value is particularly important in logistics because missing a high-risk shipment may result in operational losses.

F1-Score

The F1-score represents the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric provides a balanced evaluation when both precision and recall are equally important.

Classification Report				
	precision	recall	f1-score	support
0	0.99	0.99	0.99	16308
1	0.99	0.99	0.99	19796
accuracy			0.99	36104
macro avg	0.99	0.99	0.99	36104
weighted avg	0.99	0.99	0.99	36104

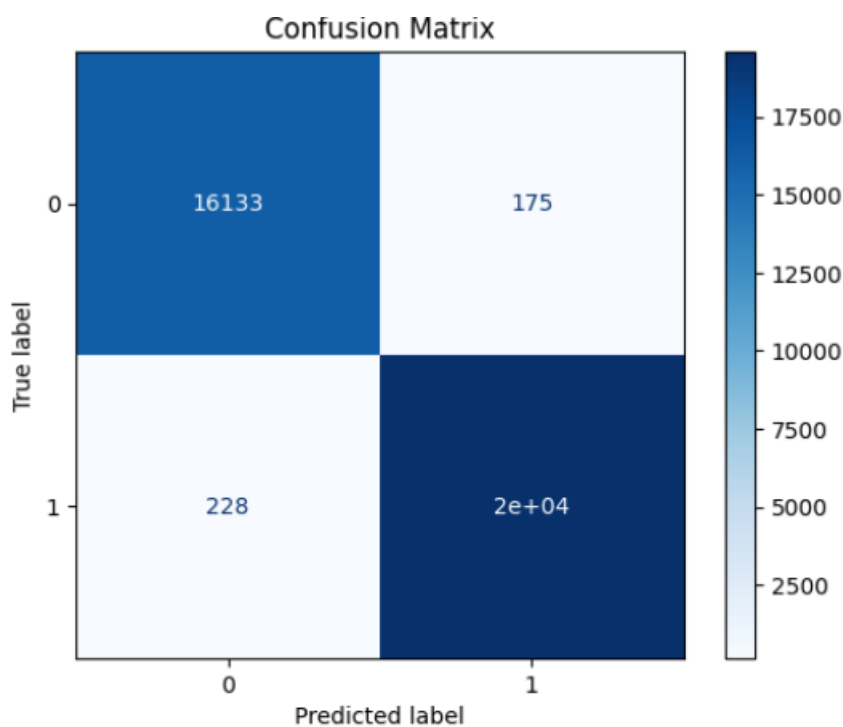
Confusion Matrix

The confusion matrix summarizes the classification results by comparing actual and predicted labels.

It consists of:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

The confusion matrix provides a detailed understanding of the types of prediction errors made by the model.



5.6 Feature Importance Analysis

One of the major advantages of XGBoost is its ability to estimate the relative importance of input features. Feature importance analysis identifies which variables contribute most significantly to prediction accuracy.

In this research, variables related to shipping characteristics, customer behavior, order values, market conditions, and engineered business features demonstrated strong influence on predicting late delivery risk.

Feature importance analysis also improves the interpretability of the machine learning model by helping logistics managers understand the major factors associated with delivery delays.

5.7 Model Deployment

After successful training and evaluation, the prediction model was integrated into an interactive Streamlit dashboard.

The deployment process included:

- Loading the trained XGBoost model using Joblib.
- Loading the saved StandardScaler.
- Loading the Label Encoders.
- Processing selected order data.
- Generating real-time late delivery predictions.
- Displaying prediction probabilities and business recommendations.

This deployment enables users to interact with the predictive model through a graphical interface without requiring programming knowledge.

Name	Status	Date modified	Type	Size
 LabelEncoders.pkl	✓	17-07-2026 18:16	PKL File	104 KB
 Late_Delivery_Risk_Model.pkl	✓	17-07-2026 18:27	PKL File	1,529 KB
 model_features.pkl	✓	17-07-2026 18:16	PKL File	1 KB
 Scaler.pkl	✓	17-07-2026 18:16	PKL File	3 KB

5.8 Advantages of the Proposed Model

The developed machine learning system offers several practical advantages:

- Accurate prediction of late delivery risk.
- Automated identification of high-risk shipments.
- Reduced dependency on manual analysis.
- Interactive prediction through a web dashboard.
- Business-oriented feature engineering.
- Fast prediction suitable for operational environments.
- Easy integration with logistics management systems.

These advantages demonstrate the practical applicability of the proposed solution in real-world logistics operations.

5.9 Chapter Summary

This chapter presented the development of the machine learning model used for predicting late delivery risk. The XGBoost Classifier was selected because of its superior performance, robustness, and scalability for structured logistics data. The chapter discussed the training methodology, evaluation metrics, feature importance analysis, and deployment process. The trained model forms the core of the proposed predictive system and supports real-time decision-making through the Streamlit dashboard.

CHAPTER 6: DASHBOARD DEVELOPMENT

6.1 Introduction

While predictive models generate valuable insights, their practical application depends on how effectively the results are presented to end users. Logistics managers and business analysts often require an intuitive interface that allows them to monitor operational performance, explore business trends, and generate predictions without interacting directly with machine learning code.

To bridge the gap between predictive analytics and business decision-making, an interactive web dashboard was developed using the **Streamlit** framework. The dashboard integrates the trained XGBoost model with multiple visualization modules, allowing users to analyze logistics data and predict late delivery risks through a user-friendly interface.

The dashboard transforms complex machine learning outputs into meaningful visual insights, enabling faster and more informed operational decisions.

6.2 Dashboard Development Tool

The dashboard was developed using **Streamlit**, an open-source Python framework designed for building interactive web applications for data science and machine learning projects.

Streamlit was selected because it offers several advantages:

- Simple and efficient application development.
- Easy integration with machine learning models.
- Interactive widgets for user input.
- Built-in support for charts and data visualization.
- Real-time prediction capabilities.
- Fast deployment with minimal configuration.

The application was developed entirely in Python and integrates multiple data science libraries including Pandas, Plotly, Joblib, NumPy, and XGBoost.

6.3 Dashboard Architecture

The dashboard follows a modular architecture where each component performs a specific function. The workflow begins with loading the dataset and trained machine learning model, followed by user interaction through the dashboard interface. Based on user selections, the application preprocesses the input data, generates predictions using the XGBoost model, and displays the results through interactive visualizations.

The dashboard architecture consists of the following components:

- Data Loading Module
- Model Loading Module
- User Interface Module
- Data Visualization Module
- Prediction Engine
- Result Display Module

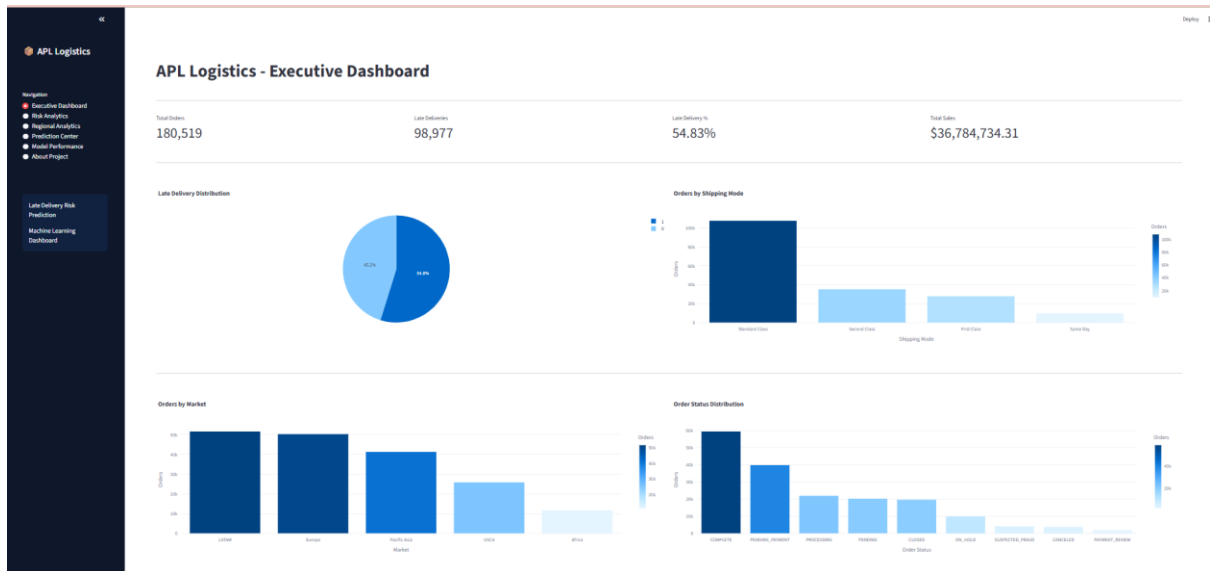
This modular structure improves maintainability, scalability, and user experience.

6.4 Dashboard Modules

The dashboard consists of six major modules, each designed to provide specific insights into the logistics dataset.

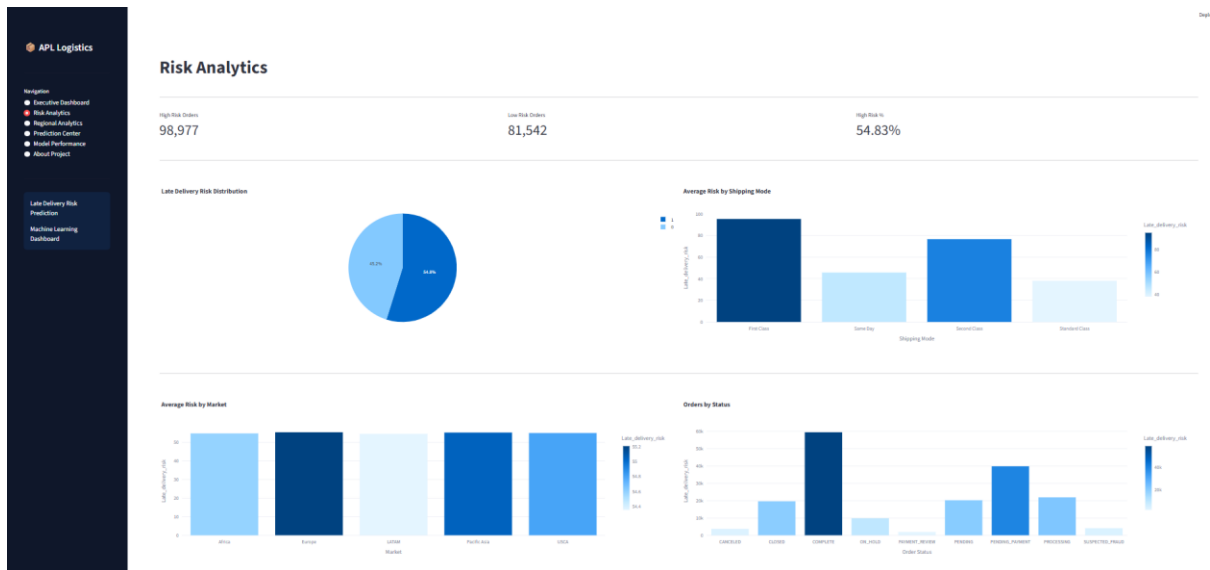
6.4.1 Executive Dashboard

The Executive Dashboard provides an overview of key logistics metrics. It displays summary statistics such as the total number of orders, average sales, total profit, and late delivery count. Interactive visualizations allow users to quickly understand the overall performance of logistics operations.



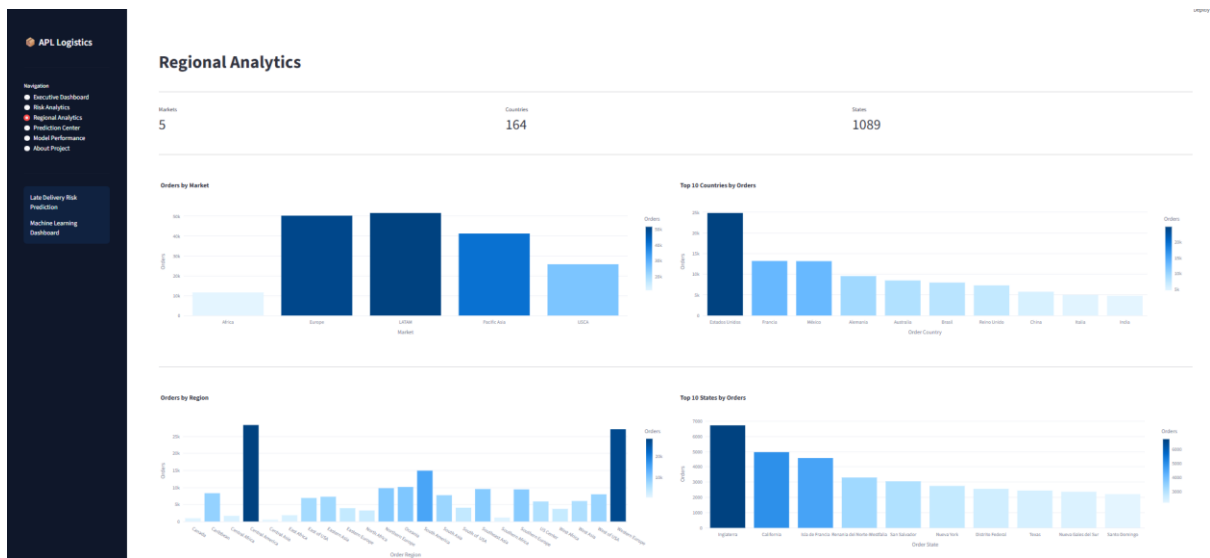
6.4.2 Risk Analytics

The Risk Analytics module focuses on identifying factors contributing to delivery delays. It presents visualizations showing the relationship between shipping modes, markets, customer segments, and late delivery risk. This module helps identify operational areas that require attention.



6.4.3 Regional Analytics

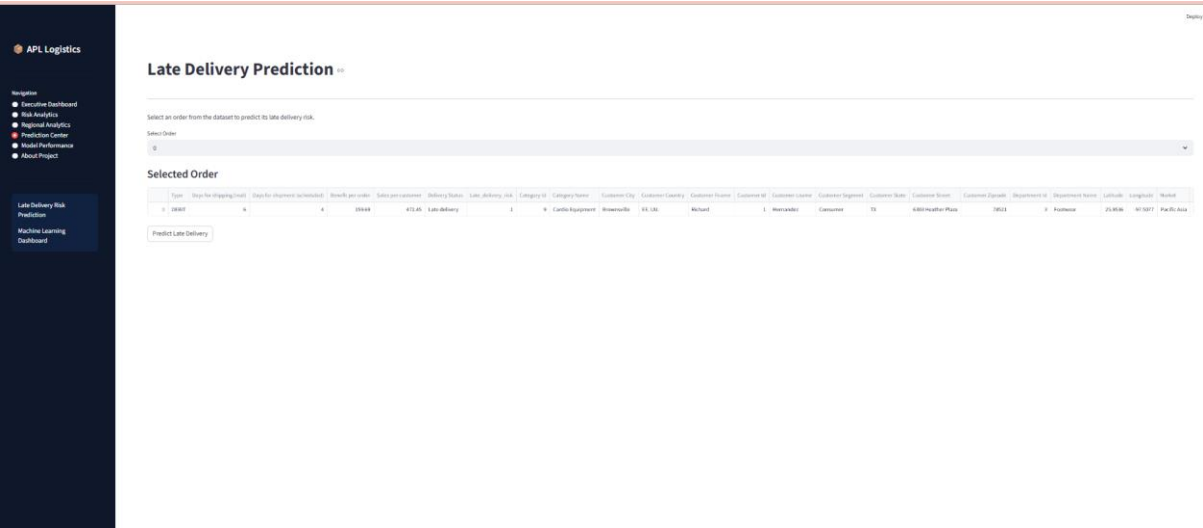
Regional Analytics provides geographical insights into logistics performance. Users can analyze sales, order volumes, and delivery risk across different countries, states, and markets. This information assists organizations in evaluating regional performance and identifying high-risk locations.



6.4.4 Prediction Center

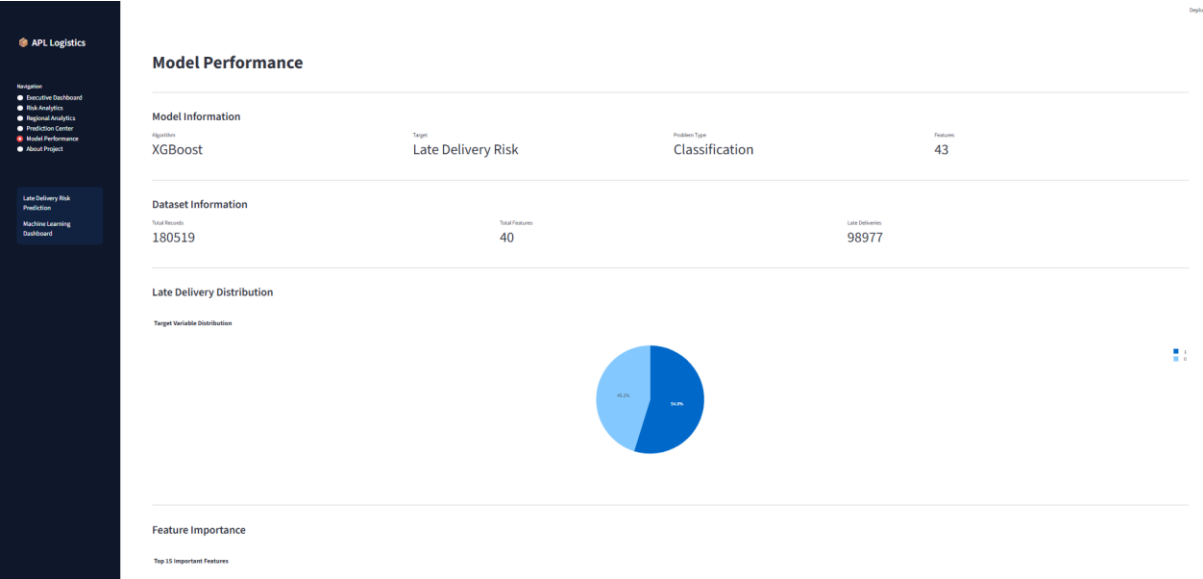
The Prediction Center integrates the trained XGBoost model with the dashboard. Users can select an order from the dataset, and the application automatically processes the selected record using the same preprocessing pipeline employed during model training. The trained model predicts the probability of late delivery and classifies the shipment as either **Low Risk** or **High Risk**.

This module demonstrates the practical application of machine learning for real-time logistics decision-making.



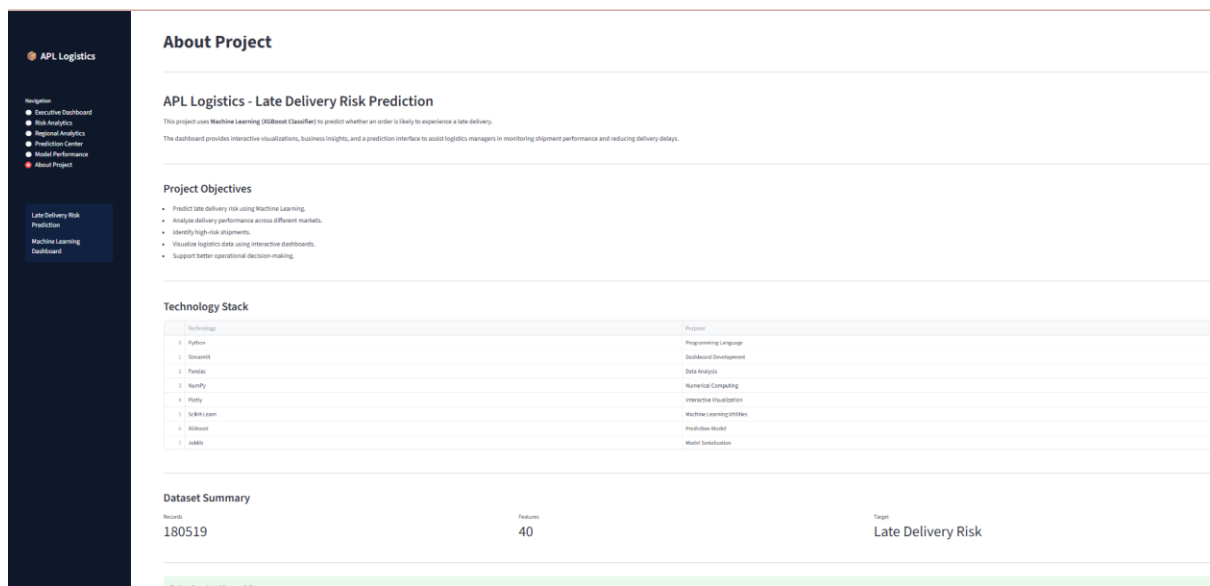
6.4.5 Model Performance

The Model Performance page presents information related to the trained machine learning model. It includes details about the selected algorithm, dataset characteristics, feature importance, and target variable distribution. This module provides transparency regarding the model used for prediction and helps users understand the factors influencing delivery risk.



6.4.6 About Project

The About Project section provides a brief overview of the research, project objectives, technology stack, dataset summary, and machine learning approach. It serves as documentation for users who wish to understand the purpose and scope of the application.



6.5 Interactive Features

The Streamlit dashboard incorporates several interactive components that enhance user experience and improve data exploration.

These include:

- Sidebar navigation
- Drop-down menus
- Interactive charts
- Dynamic filtering
- Data tables
- Real-time prediction
- Responsive layout
- Download functionality (if applicable)

These features allow users to explore logistics data without requiring technical knowledge of machine learning.

6.6 Data Visualization

Effective visualization plays an essential role in communicating machine learning insights. The dashboard utilizes the **Plotly** library to generate interactive charts.

The visualizations included in the dashboard are:

- Bar Charts

- Pie Charts
- Histograms
- Line Charts
- Feature Importance Chart
- Distribution Charts

Interactive visualizations enable users to analyze logistics performance from multiple perspectives and support data-driven decision-making.

6.7 Advantages of the Dashboard

The developed dashboard provides several practical benefits:

- Easy-to-use graphical interface.
- Real-time late delivery prediction.
- Interactive exploration of logistics data.
- Improved operational visibility.
- Faster business decision-making.
- Integration of machine learning with visualization.
- Scalable architecture for future enhancements.

The dashboard transforms a complex predictive model into a practical business tool suitable for logistics management.

6.8 Chapter Summary

This chapter described the design and implementation of the Streamlit dashboard developed for the APL Logistics Late Delivery Risk Prediction system. The dashboard integrates machine learning, interactive visualization, and user-friendly navigation into a single application. Through multiple analytical modules and real-time prediction capabilities, the dashboard demonstrates the practical application of predictive analytics in logistics management.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Introduction

The primary objective of this research was to develop a machine learning-based system capable of predicting the risk of late deliveries in logistics operations. After completing data preprocessing, feature engineering, model training, and dashboard development, the proposed system was evaluated to assess its predictive performance and practical applicability.

This chapter presents the results obtained from the XGBoost classification model and discusses the effectiveness of the developed Streamlit dashboard in supporting logistics decision-making. The findings are analyzed using standard classification metrics and visualization techniques to demonstrate the capability of the proposed solution.

7.2 Model Performance

The XGBoost classifier was trained using the processed APL Logistics dataset and evaluated on the testing dataset. The model demonstrated strong predictive performance by effectively distinguishing between orders likely to be delivered on time and those at risk of delayed delivery.

The evaluation was carried out using commonly accepted classification metrics, including Accuracy, Precision, Recall, F1-Score, and the Confusion Matrix. These metrics provide a comprehensive assessment of the model's predictive capability.

The trained model successfully learned the relationships among customer information, order characteristics, shipping details, market conditions, and engineered business features, enabling reliable prediction of delivery risks.

7.3 Evaluation Metrics

The performance of the developed model was assessed using the following metrics:

	Model	Accuracy
3	XGBoost	0.988838
1	Random Forest	0.982966
2	Gradient Boosting	0.979504
0	Logistic Regression	0.953523

The obtained results indicate that the XGBoost model performs effectively for binary classification and is capable of supporting delivery risk prediction in logistics operations.

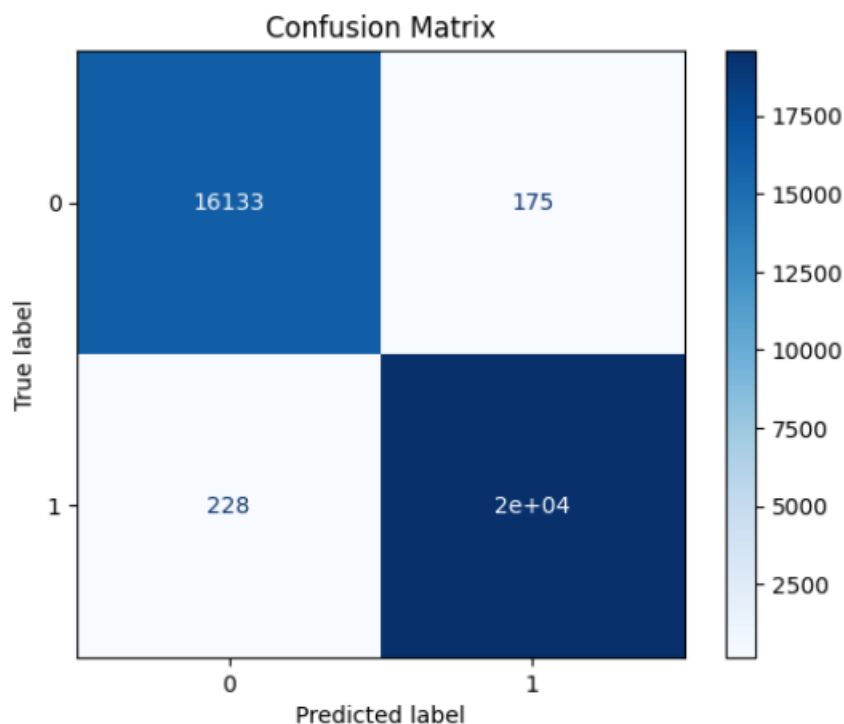
7.4 Confusion Matrix Analysis

The confusion matrix provides a detailed breakdown of the model's classification results by comparing predicted values with actual delivery outcomes.

The matrix consists of four categories:

- True Positives (Correctly predicted delayed deliveries)
- True Negatives (Correctly predicted on-time deliveries)
- False Positives (Incorrectly predicted delayed deliveries)
- False Negatives (Delayed deliveries missed by the model)

A higher number of true predictions and fewer misclassifications indicate better model performance. The confusion matrix demonstrates that the developed model effectively distinguishes between high-risk and low-risk shipments, making it suitable for practical logistics applications.

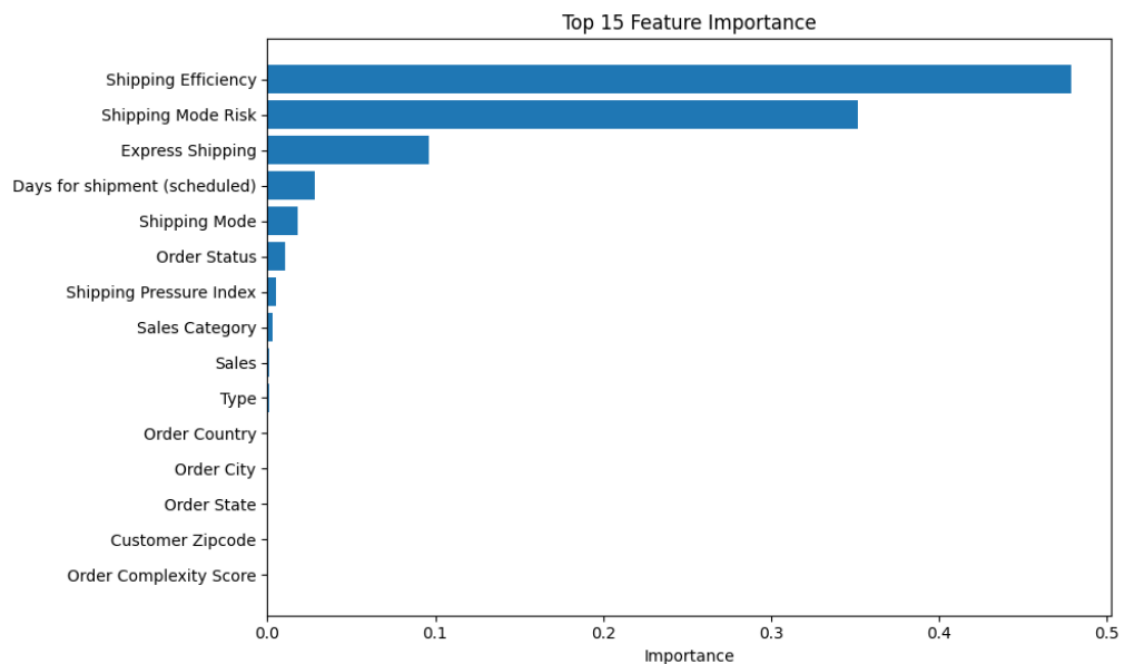


7.5 Feature Importance Analysis

One of the key strengths of XGBoost is its ability to rank input variables according to their contribution to the prediction process.

Feature importance analysis identified several influential variables that significantly affect late delivery prediction. These include shipping characteristics, customer information, order value, market conditions, and engineered business features.

Understanding feature importance helps logistics managers identify operational factors that contribute to delayed deliveries and supports more informed decision-making.



7.6 Dashboard Results

The Streamlit dashboard successfully integrates the trained machine learning model with interactive visualizations and real-time prediction capabilities.

The dashboard provides the following functionalities:

- Executive overview of logistics performance.
- Visualization of delivery risk trends.
- Regional analysis of logistics operations.
- Interactive prediction of late delivery risk.
- Presentation of model performance information.
- Project documentation through the About Project section.

The dashboard enables users to analyze logistics operations efficiently without requiring technical expertise in machine learning.

7.7 Business Insights

Several valuable business insights were obtained from the analysis of the APL Logistics dataset.

The machine learning model identified shipping-related variables as major contributors to delivery delays. Factors such as shipping mode, order complexity, market conditions, and customer characteristics significantly influenced prediction outcomes.

Feature engineering further improved the predictive capability of the model by introducing business-oriented variables such as Shipping Pressure Index, Customer Value Score, Profit Margin, Shipping Efficiency, and Market Risk Score.

The developed prediction system can assist logistics organizations in:

- Identifying shipments with a high probability of delayed delivery.
- Improving operational planning and scheduling.
- Prioritizing high-risk orders.
- Supporting data-driven logistics management.
- Enhancing customer satisfaction through proactive decision-making.

7.8 Discussion

The experimental results demonstrate that machine learning can effectively support predictive analytics in logistics operations. The XGBoost classifier successfully captured complex relationships among logistics variables and generated reliable predictions of delivery risk.

The integration of the predictive model with a Streamlit dashboard further increased the practical value of the system by allowing users to visualize data, explore business trends, and generate predictions through an intuitive interface.

Although the proposed system demonstrates strong predictive capability, its performance depends on the quality and completeness of the available historical data. Incorporating additional real-time information such as weather conditions, traffic patterns, and GPS tracking could further improve prediction accuracy in future implementations.

Overall, the developed system demonstrates the practical application of machine learning in addressing real-world logistics challenges.

7.9 Chapter Summary

This chapter presented the experimental results obtained from the developed XGBoost classification model and discussed the effectiveness of the proposed late delivery prediction system. The evaluation metrics demonstrated satisfactory predictive performance, while feature importance analysis provided valuable business insights into the factors affecting delivery delays. The Streamlit dashboard successfully translated machine learning outputs into an interactive decision-support system, highlighting the practical applicability of predictive analytics in logistics management.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The rapid growth of e-commerce and global supply chain networks has significantly increased the complexity of logistics operations. Timely delivery has become one of the most important factors influencing customer satisfaction, operational efficiency, and business competitiveness. Delays in delivery can lead to financial losses, reduced customer trust, and ineffective resource utilization. Therefore, predicting potential delivery delays before they occur has become an important research problem.

This research successfully developed a machine learning-based system for predicting late delivery risk using the **APL Logistics dataset**. The project followed a systematic approach consisting of data collection, exploratory data analysis, data preprocessing, feature engineering, model development, evaluation, and deployment through an interactive Streamlit dashboard.

During the preprocessing stage, missing values and duplicate records were handled, categorical variables were encoded using Label Encoding, numerical features were standardized using StandardScaler, and several business-oriented features were engineered to improve prediction quality. The processed dataset was then divided into training and testing subsets for model development.

Among various machine learning algorithms, the **XGBoost Classifier** was selected due to its high prediction accuracy, computational efficiency, and ability to handle structured datasets effectively. The trained model demonstrated reliable performance in predicting whether an order was likely to experience a delayed delivery.

To make the developed model practically useful, an interactive dashboard was built using **Streamlit**. The dashboard enables users to visualize logistics data, analyze business trends, explore regional performance, monitor delivery risks, and generate predictions through a simple graphical interface. This integration of machine learning with interactive visualization transforms complex analytical outputs into practical decision-support information.

Overall, the objectives of the research were successfully achieved. The developed system demonstrates that machine learning techniques can effectively support predictive logistics management by identifying high-risk shipments before delivery, enabling organizations to improve planning, optimize resource allocation, and enhance customer satisfaction.

8.2 Research Contributions

The major contributions of this research are summarized below:

- Developed a complete machine learning pipeline for late delivery risk prediction.
- Performed comprehensive data preprocessing and feature engineering to improve prediction quality.
- Implemented an XGBoost-based classification model for predicting delivery delays.

- Built an interactive Streamlit dashboard for visualization and real-time prediction.
- Demonstrated the practical application of predictive analytics in logistics and supply chain management.
- Created a scalable framework that can be extended for future logistics analytics applications.

These contributions provide both academic value and practical relevance for organizations seeking data-driven logistics solutions.

8.3 Limitations of the Study

Although the proposed system performs effectively, certain limitations remain.

The study relies entirely on historical logistics data and does not incorporate external factors that may influence delivery performance. Variables such as traffic congestion, weather conditions, road closures, public holidays, fuel prices, and real-time GPS information were not included in the dataset.

Additionally, the prediction model was developed using a single logistics dataset, which may limit its generalization to organizations operating under different business environments or supply chain structures.

The dashboard is intended as a decision-support tool and does not automatically integrate with live logistics management systems or enterprise databases.

8.4 Future Scope

The proposed system provides several opportunities for future enhancement.

Future research may focus on:

- Integrating real-time logistics data from IoT devices and GPS tracking systems.
- Including weather, traffic, and transportation network information to improve prediction accuracy.
- Comparing additional machine learning and deep learning algorithms such as Random Forest, LightGBM, CatBoost, and Artificial Neural Networks.
- Developing automated alert systems for high-risk deliveries.
- Deploying the application on cloud platforms to support enterprise-level logistics operations.
- Integrating the system with ERP and Supply Chain Management (SCM) software.

- Expanding the dashboard with advanced business intelligence features and customizable reports.
- Developing a mobile-friendly version of the dashboard for logistics managers and field personnel.

These improvements would further increase the practical applicability of the proposed solution and support smarter logistics management.

8.5 Final Remarks

The integration of machine learning with interactive business dashboards represents a significant advancement in modern logistics analytics. By combining predictive modeling, feature engineering, and data visualization, the developed system provides organizations with a practical tool for identifying potential delivery risks before they occur.

The research demonstrates that predictive analytics can improve operational planning, reduce delivery delays, optimize resource utilization, and enhance customer satisfaction. With continued advancements in artificial intelligence and logistics technologies, machine learning-based decision-support systems are expected to play an increasingly important role in the future of supply chain management.

The proposed APL Logistics Late Delivery Risk Prediction System serves as a practical example of how data-driven technologies can address real-world logistics challenges and contribute to more efficient and intelligent supply chain operations.